

Research Interests

Gradient flows and optimal transport; metric-space evolution equations and interacting populations; game theory, strategic learning, and control.

Academic Appointments

- Sept. 2026–
present **Stanford University**
Szegő Assistant Professor, Department of Mathematics
Mentors: Lenya Ryzhik and Lexing Ying.
- July–Aug.
2026 **California Institute of Technology**
Postdoctoral Research Fellow
Mentor: Franca Hoffmann.

Education

- 2026 **California Institute of Technology**
Ph.D. in Control and Dynamical Systems
Advisors: Franca Hoffmann, Eric Mazumdar, and John Doyle.
- 2018 **Cornell University**
B.S. in Electrical and Computer Engineering, *summa cum laude*.

Honors and Funding

- 2026 NSF Mathematical Sciences Postdoctoral Research Fellowship
2026 Ben P.C. Chou Doctoral Prize in Information Science and Technology
2026 US Junior Oberwolfach Fellows NSF Grant
2025 Best Student Paper Award, American Control Conference
2023 PIMCO Graduate Fellowship in Data Science
2022 NDSEG Graduate Research Fellowship Awardee
2022 NSF Graduate Research Fellowship Awardee
2021 Linde Institute of Economic Sciences Research Grant

Publications and Preprints

† Undergraduate advisee.

Preprints

- [1] **L. Conger**, F. Hoffmann, and G. Savaré. *Dissipative Evolutions in Metric Spaces*. Preprint.
- [2] **L. Conger**, F. Hoffmann, E. Mazumdar, and L. J. Ratliff. *Coupled Wasserstein Gradient Flows for Min-Max and Cooperative Games*. Under review.

Journal Publications

- [3] **L. Conger**, F. Hoffmann, E. Mazumdar, and L. J. Ratliff. *Monotone Multispecies Flows*. *SIAM Dynamical Systems*, 2026.
- [4] H. Atakisi, **L. Conger**, D. W. Moreau, and R. E. Thorne. *Resolution and dose dependence of radiation damage in biomolecular systems*. *International Union of Crystallography Journal*, vol. 6, part 6, pp. 1040–1053, 2019.

Publications and Preprints (continued)

Conference Publications

- [5] **L. Conger**, F. Hoffmann, R. Baptista, and E. Mazumdar. *Computing Optimal Transport Plans via Min-Max Gradient Flows*. IEEE Conference on Decision and Control (CDC), 2025.
- [6] **L. Conger**, A. Leeman, and F. Hoffmann. *Convex Constrained Controller Synthesis for Evolution Equations*. American Control Conference (ACC), 2025. **Best Student Paper Award**.
- [7] **L. Conger**, Y. Lin, E. Mazumdar, and A. Wierman. *Characterizing Controllability and Observability for Systems with Locality, Communication, and Actuation Constraints*. CDC, 2024.
- [8] Y. Li, J. Yu, **L. Conger**, T. Kargin, and A. Wierman. *Learning the Uncertainty Sets for Control Dynamics via Set Membership: A Non-Asymptotic Analysis*. International Conference on Machine Learning (ICML), 2024.
- [9] **L. Conger**, F. Hoffmann, E. Mazumdar, and L. Ratliff. *Strategic Distribution Shift of Interacting Agents via Coupled Gradient Flows*. Neural Information Processing Systems (NeurIPS), 2023.
- [10] **L. Conger**, S. Vernon[†], and E. Mazumdar. *Designing System Level Synthesis Controllers for Nonlinear Systems with Stability Guarantees*. Learning for Decision and Control (L4DC), 2023.
- [11] **L. Conger**, J. S. Li, E. Mazumdar, and S. L. Brunton. *Nonlinear System Level Synthesis for Polynomial Dynamics*. CDC, 2022.
- [12] **L. Conger** and S.-H. Tseng. *Output-Feedback System Level Synthesis via Dynamic Programming*. ACC, 2022.

Workshop Papers

- [13] **L. Conger**, F. Hoffmann, E. Mazumdar, and L. J. Ratliff. *Coupled Gradient Flows for Strategic Non-Local Distribution Shift*. ICLR Workshop on New Frontiers in Learning, Control, and Dynamical Systems, 2023. Full version: [9].
- [14] A. Gnad, J. Belarge, A. Canciani, **L. Conger**, J. Curro, A. Edelman, P. Morales, M. O’Keeffe, J. Taylor, and C. Rackauckas. *Signal Enhancement for Magnetic Navigation Challenge Problem*. JuliaCon, 2020.

Conferences, Schools, and Visits

* Host funding † Fellowship funding

- Aug. 2026 Workshop on Partial Differential Equations for Many Particles Systems*. ICERM Providence, USA.
- July 2026 FoCM Workshop Computational Optimal Transport*. Vienna, Austria.
- July 2026 Prof. Jan Maas. Vienna, Austria.
- Mar. 2026 Oberwolfach Workshop: Flows on Measure Spaces and Applications in Machine Learning*. Oberwolfach-Walke, Germany.
- Mar. 2026 Nonlinear Partial Differential Equations Spring School*. Ulm, Germany.
- Nov. 2025 Prof. Giuseppe Savaré, research visit. Milan, Italy.
- Aug. 2025 Mathematics of Adversarial Machine Learning*. Banff International Research Station, Oaxaca, Mexico.
- July 2025 IEEE American Control Conference*. Denver, USA.
- Apr. 2025 Workshop on Dynamics of Density Operators. IPAM Los Angeles, USA.
- July 2025 Wasserstein Gradient Flows in Math and Machine Learning*. Banff International Research Station, Banff, Canada.
- Dec. 2024 IEEE Conference on Decision and Control^{†,*}. Milan, Italy.
- Dec. 2024 Oberwolfach Mini-Workshop: High-Dimensional Control Problems and Mean-Field Equations with Applications in Machine Learning^{†,*}. Oberwolfach-Walke, Germany.

- July 2024 Theory of Interacting Particle Systems Mini-Symposium, European Congress of Mathematics*. Seville, Spain.
- July 2024 Modern Perspectives in Applied Mathematics, Theory and Numerics of PDEs. Zürich, Switzerland.
- July 2024 Prof. André Schlichting, research visit*. Münster, Germany.
- July 2024 Frontiers in Interacting Particle Systems, Aggregation-Diffusion Equations & Collective Behavior Summer Research School*. CIRM Marseille, France.
- May 2024 Interacting Particle Systems: Analysis, Control, Learning and Computation*. ICERM Providence, USA.
- Apr. 2024 International Conference on Multiscale Modeling and Simulation based on Physics and Data. IPAM Los Angeles, USA.
- Apr. 2024 Aggregation-Diffusion Equations & Collective Behavior: Analysis, Numerics and Applications Conference*. CIRM Marseille, France.
- Dec. 2023 Neural Information Processing Systems (NeurIPS). New Orleans, USA.
- July 2023 International Conference on Machine Learning (ICML). Honolulu, USA.
- June 2023 Learning for Decision and Control Conference (L4DC)[†]. Philadelphia, USA.
- Dec. 2022 IEEE Conference on Decision and Control[†]. Cancun, Mexico.
- June 2022 IEEE American Control Conference. Atlanta, USA.

Earlier Research and Industry Experience

- Feb. 2019–**MIT Lincoln Laboratory**, Assistant Technical Staff.
- Aug. 2020 Radar simulation, tracking and learning algorithms, high-dimensional optimization, and magnetic-signal denoising.
- Summers **Raytheon Missile Systems**, internships.
- 2016–2018 Signal Processing Intern (2017, 2018); Modeling, Simulation & Analysis Intern (2016). Signal processing, Kalman filtering, simulation, and data-analysis tools in MATLAB, Simulink, and Python.

Selected Talks

- Dissipative Evolutions in Metric Spaces*
- Oct. 2026 Stanford Applied Math Seminar
- July 2026 FoCM Workshop
- Games & Gradient Flows: Modeling Strategic Behavior*
- Mar. 2026 Oberwolfach Workshop: Flows on Measure Spaces and Applications in Machine Learning
- Monotonicity of Coupled Multispecies Wasserstein-2 Gradient Flows*
- Mar. 2026 Nonlinear Partial Differential Equations Spring School
- Aug. 2025 BIRS Oaxaca: Mathematical Analysis of Adversarial Machine Learning
- July 2025 BIRS: Wasserstein Gradient Flows in Math and Machine Learning
- July 2025 University of Washington, Data-Driven Seminar
- A New Control Parameterization for Constrained Systems*
- Dec. 2024 Oberwolfach Mini-Workshop: High-Dimensional Control Problems and Mean-Field Equations with Applications in Machine Learning
- Convex Constrained Controller Synthesis for Evolution Equations*
- July 2024 University of California San Diego, CoPI Seminar

Strategic Distribution Shift of Interacting Agents via Coupled Gradient Flows

- Sept. 2024 University of Southern California, CSC@USC Seminar Series
- July 2024 European Congress of Mathematics, Theory of Interacting Particle Systems Mini-Symposium
- July 2024 ETH Zurich, Intelligent Control Systems Lab Seminar
- July 2024 Universität Münster, Kolloquium der angewandten Mathematik
- Apr. 2024 UCLA, SoCal Control Workshop
- Dec. 2023 Berkeley, Semiautonomous Seminar
- May 2023 Harvard College, Group Seminar
- May 2023 MIT Lincoln Laboratory, Homeland Protection Systems Research Seminar

Teaching Experience

California Institute of Technology

- Winter 2025 **Teaching Assistant and Lecturer:** CDS 131, Linear Systems
Teaching Assistant
- Winter 2026 ACM 107b: Functional Analysis
- Fall 2025 CS 143: Networks: Algorithms & Architecture
- Winter 2024 CS 42: Computer Science Education in K–14 Settings
- Fall 2023 CDS 131: Linear Systems
- Winter 2021 CS 144: Networks and Economics

Cornell University

Teaching Assistant

- Spring 2017; Fall 2018 PHYS 2214: Oscillations, Waves, and Quantum Physics
- Spring 2018 ECE 3250: Mathematics of Signals and Systems
- Fall 2016 PHYS 1112: Mechanics and Heat
- Fall 2018 **Grader:** ECE 2200, Signals and Information

Mentorship

- 2025 **Minh Nguyen** (Caltech Ph.D. student)
Particle method for Knothe–Rosenblatt optimal transport maps.
- 2023–2024 **Katherine Graham** (University of Alabama Birmingham)
Data analysis of health care outcomes and insurance coverage.
- 2023 **Sarvagna Velidandla** (Caltech)
Data analysis of the effects of COVID-19 on high school athletes.
- 2022 **Sydney Vernon** (Caltech)
Machine learning applications of nonlinear control; coauthor on L4DC 2023 paper. Next step: UW Research Assistant.
- 2022–2023 **Graduate application mentoring**, Caltech Accountability Partners Program: Ting Li (Johns Hopkins), 2023; Sultan Daniels (Brown), 2022 (next step: Berkeley EECS Ph.D. student).

Professional Service

Seminar Organization

- 2025–2026 **Gradient Flow Reading Group**, organizer.
Discussions for Ph.D. students and postdocs; 20 discussion leaders from the United States and Europe.
- 2024–2025 **Hoffmann Group Seminars**, organizer.
Guest and internal talks on PDE analysis, mean field limits, and related fields.

Reviewing

Journal: IEEE Transactions on Automatic Control (2023, 2026).
Conferences: ACC and CDC (2022–2025); L4DC (2023).
Workshop: ICML New Frontiers in Learning, Control, and Dynamical Systems (2023).

Departmental Service

- 2024–2025 Student Member, Caltech CMS AI/ML Admissions Committee.
Reviewed 30+ graduate applications and made recommendations to faculty.
- 2021–2024 Computing & Mathematical Sciences Advisory Board
- 2023 Women in CMS, organizer

Outreach

- 2023–2025 **Family Promise**, coordinator and host.
Hosted families experiencing homelessness and coordinated graduate-student volunteers, including wildfire response.
- 2023–2024 **Pasadena Public Schools**, Science Night volunteer.
Designed and organized hands-on computer science activities for elementary students.
- 2020–2024 **Southern AZ Science Fair**, middle school judge and blog writer
- 2021 **iSTEM Scholars**, leader of a four-person summer mentor team

References

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